

Natural Language Processing: Attention Mechanisms, Tokenization, and Contextual Embeddings

1. Abstract & Introduction

Natural Language Processing (NLP) has been revolutionized by self-attention mechanisms and Transformer architectures. This paper investigates contextual word embeddings (BERT, GPT), subword tokenization (Byte-Pair Encoding), and sequence-to-sequence translation.

2. Attention Mechanics & Named Entity Recognition

Multi-head attention computes Query, Key, and Value matrix projections to capture long-range linguistic dependencies. Named Entity Recognition (NER) and sentiment classification tasks demonstrate state-of-the-art accuracy when fine-tuned on domain specific text.

3. Results

Fine-tuned Transformer models reached a BLEU score of 41.2 on machine translation and an F1-score of 92.1% on domain specific NER tasks.

4. Conclusion

Self-attention mechanisms provide superior context modeling compared to traditional recurrent architectures (LSTM/GRU).